

MRISA: A Visual Analytics Approach for Comparing Locomotion Policies in Robotics Training

Nan Li

nan.li@uzh.ch
University of Zurich
Zurich, Switzerland

Xiaoyu Zhang*

xiaoyu.zhang@cityu.edu.hk
City University of Hong Kong
Hong Kong, China

Fan Shi

fan.shi@nus.edu.sg
National University of Singapore
Singapore

April Yi Wang*

april.wang@inf.ethz.ch
ETH Zurich
Zurich, Switzerland

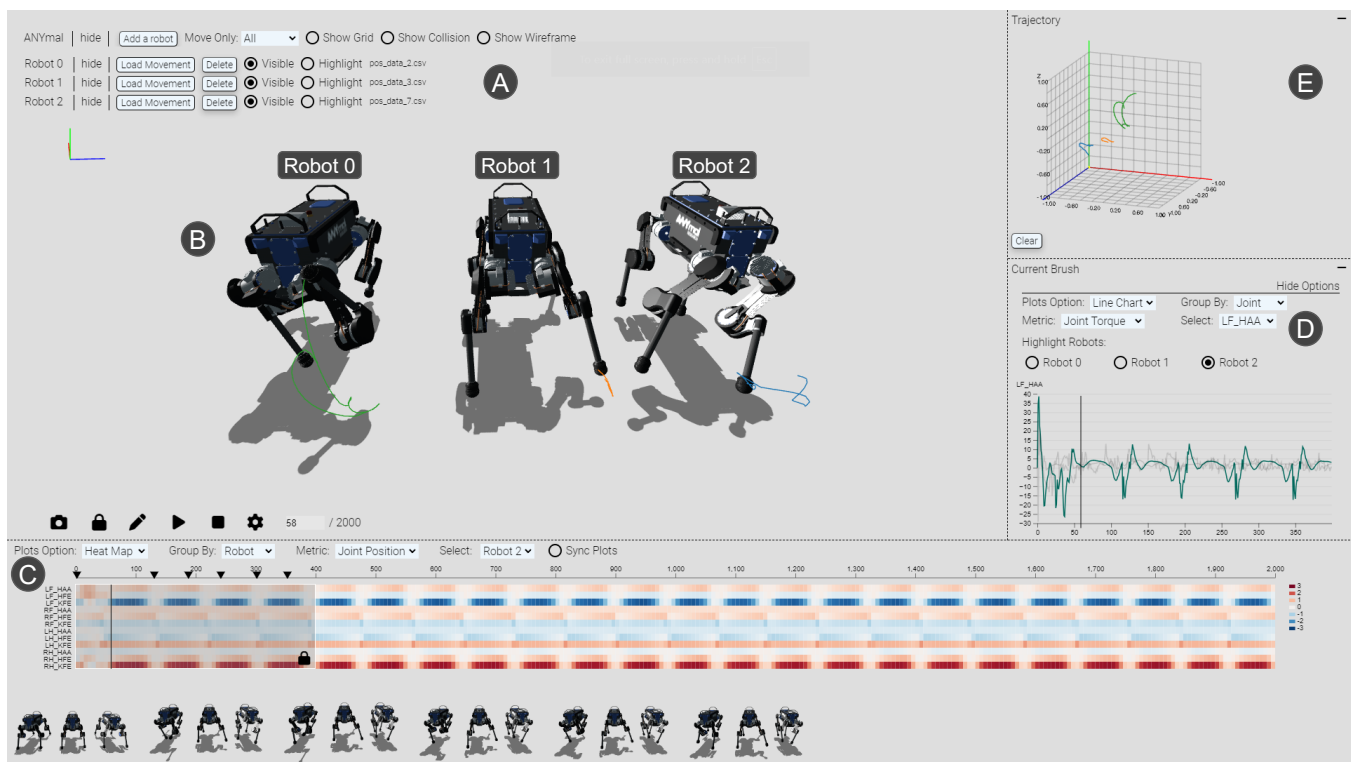


Figure 1: MRISA Interface. (A) Top Control Panel: Control options that allow users to add, hide, highlight, and load specific policy data for robots, along with simulator settings. (B) Simulation Panel: Simulation of three quadruped robots ANYmal [27], showing their current positions, rotations, and movement trajectories for one leg. (C) Global Time-based Data View: Global timeline-based data providing an overview of the entire loaded dataset, with snapshots of keyframes and the corresponding steps shown on the timeline. (D) Local Data View: Local data view enabling detailed analysis within a selected range, defined by the brush tool applied to the global timeline in Global Time-based Data View. (E) Trajectory Coordinate Panel: 3D trajectory plot with a coordinate system, visualizing the movement paths to scale.

*Corresponding Authors

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Abstract

In the field of robotics, the process of locomotion control policy training is inherently iterative and exploratory. Practitioners often switch between multiple simulation and data analysis tools to observe robot postures and behaviors, track part movements, and compare reward data, which is a tedious process. To better understand their hurdles and requirements, we interviewed five robotics experts and analyzed representative figures from recent robotics publications to identify prevailing challenges and strategies for comparing and communicating locomotion policies. The main challenges include the lack of integrated simulation and visualization, difficulty in comparing multiple policies simultaneously, and the time-intensive process of creating polished visual presentations. Based on the insights, we introduce *MRISA* (Multi-Robot Interactive Simulation and Analysis Platform), an interactive tool designed to support exploratory analysis on pre-trained locomotion policies. *MRISA* integrates features including direct observation of one or multiple robots' behaviors in a simulator, trajectories visualization with customized anchors, key measurement inspection in a timeline view, and key frames capturing. A user evaluation with 14 domain practitioners demonstrated that *MRISA* provides immediate insights, enabling practitioners to intuitively explore multiple dimensions of locomotion policies.

CCS Concepts

• **Human-centered computing** → **Visualization systems and tools**; **Interactive systems and tools**.

Keywords

Interactive Machine Learning, Robotics, Visual Analytics, Model Comparison

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1 Introduction

Recent advances in Reinforcement Learning (RL) have significantly enhanced the performance of robotic control policies across a range of platforms, such as traversing diverse terrains with quadrupedal robots [39] and solving Rubik's cube with robotic hands [43]. Beyond these sophisticated tasks, RL-driven robots are also being applied to tasks in everyday life such as inspection automation [16], and household robotics [1]. These breakthroughs highlight the versatility of RL-based robots to learn and adapt to new tasks through self-improvement and interaction with their environment, showcasing the growing potential of RL in enabling more autonomous and intelligent robotic systems.

The outstanding performance is often achieved by training locomotion policies in simulated environments using tools that visualize the robot's movements. By observing simulations, practitioners can iteratively adjust training parameters like reward structures to optimize performance. Practitioners also often analyze numerical metrics to gain deeper, qualitative insights that complement



Figure 2: A well-crafted figure shows that one policy is vulnerable to instability from adversarial attacks on its observations [49], which took three hours to composite.

the insights from simulations. Despite available simulation tools [15, 28, 37, 52] and data visualization methods [17, 26], policy training using RL remains an iterative and complex process. While some simulation tools offer movement visualization, practitioners often need to write custom codes to plot or analyze the numerical data. This process forces them to switch between simulation and data analysis tools, making the workflow tedious.

Therefore, the goal of our research is to help robotics domain practitioners better analyze and compare locomotion policies with reducing redundant work during the training and communication process. We conducted interviews with five roboticists to uncover key pain points in their workflows and analyzed 58 figures from 20 leading robotics papers to identify common visualization practices. We found that keyframes with trajectories are frequently used to depict movement, while visualizations of mathematical data play a crucial role in understanding performance. Although existing tools assist practitioners in training locomotion policies, they have limited capacity to effectively evaluate and compare different policies. Additionally, creating high-quality visual displays for formal presentations, such as Figure 2, requires substantial effort and time. The findings suggest that a tool designed to offer more convenient methods for comparing different policies and generating visual displays would be highly beneficial.

We propose *MRISA* (Multi-Robot Interactive Simulation and Analysis Platform)¹, which supports coordinated visual analytics of diverse metadata with varying measurements, dimensions, and granularities, generated across different stages of policy training. *MRISA* allows users to visualize robot movements with trajectory, compare multiple policy metrics, and snapshot keyframes for easier communication. With its intuitive interface and interactive features, *MRISA* reduces the need to switch between tools, allowing users to focus on policy refinement and professional result generation, thus making the research process more efficient.

¹<https://github.com/linan1109/MIRSA>

To assess the utility and performance of *MRISA*, a user evaluation was conducted with 14 domain practitioners and demonstrated that *MRISA* effectively enables users to analyze, evaluate, and compare policies interactively, with participants giving it a high satisfaction rating. They highlighted the platform’s intuitive interface and its capability to visualize data in ways that were not possible with their previous tools. As one participant remarked, “*It solved some problems that had troubled me for a long time.*”

In summary, our work contributes:

- A **formative study** with 5 roboticists that point out the current challenges in the common workflows when training locomotion policies.
- A **content analysis** of 20 leading robotics papers, providing insights into the types of visualizations commonly used in the field.
- A prototype system, *MRISA*, which offers simulation and visualization to facilitate the comparison of multiple robots’ performance.
- A **user evaluation** that offers initial insights.

2 Related Work

Our project aims to enhance the efficiency and simplicity of robotics policy training. To contextualize this work, we review three key areas: first, existing technologies and tools for robotics policy training; second, recent advances in explainable AI for robotics; and third, interactive methods aimed at optimizing machine learning models.

2.1 Tools in Robotics Policy Training

A crucial aspect of policy learning involves not only defining the input (observation space) and output (action space) of the control policy but also crafting an appropriate reward function. The design of this reward function plays a pivotal role in shaping desired robotic behaviors and is highly task-specific, often requiring domain expertise to adjust reward terms effectively [44].

In practice, most policies are first trained in simulated environments before being deployed on physical hardware. Popular simulators include GPU-accelerated platforms such as NVIDIA Isaac [37] and multi-threaded CPU-based simulators like RaiSim [28], PyBullet [15], and MuJoCo [52]. These tools enable developers to visualize and iterate on policy performance during the simulation phase. By monitoring the behavior of the policy, developers adjust various parameters of the training process, including reward weights and hyperparameters of the learning algorithms. Once the policy demonstrates satisfactory performance in simulation, it is then considered for deployment on real-world robotic systems [29].

When it comes to visualization, simulators differ significantly in their capabilities and usability. NVIDIA Isaac, for instance, offers pop-up plots to monitor performance, but managing these over a moving time horizon—such as playing forward or backward—can be challenging. GAZEBO [47] primarily focuses on visualizing the robot within its environment, while third-party tools integrated with Robotic Operating System (ROS) [50] can provide more advanced plotting functionality through command-line interfaces. PyBullet offers similar visualization capabilities. Raisim, in contrast, provides visualization for joint information, but its fixed options

limit the user’s ability to customize other aspects of the display. In robotics policy training, there is a growing need for more flexible platforms that can support a broader range of visualization functions, enabling developers to more effectively analyze and refine their policies across diverse tasks and environments.

2.2 Explainable AI for Robotics

RL models, like many other machine learning models, are often regarded as black-box systems due to their opacity and difficulty in interpretation. Different RL models used for robot training are typically explained through various methods, most of which are technical. Some explainable RL models are proposed for robot control, such as for Unmanned Aerial Vehicles (UAVs) [20, 54], or general robot control [5].

Another approach is to use easier and more interpretable models to explain the behavior of trained RL agents. For instance, TripleTree model explains by structuring the decision-making processes of black-box agents into an interpretable tree-like representation [4]. Similarly, PW-Net abstracts the complex state space into simplified prototypes that align with human understanding and explains the policy via those prototypes [32]. RL models can also integrate a causal lens to generate more interpretable explanations for their actions based on cause-and-effect reasoning [36]. Adding perturb to the input and observing the output changes explains by highlighting the important input features [18, 30].

Visualization methods can also be used to explain RL, but mostly for image-based inputs. In Deep Q-networks (DQN), each state is reduced in lower dimension and represented as a point in the t-SNE map, which helps in visualizing the structure and relationships between different states [61]. Saliency maps highlight the areas of input that most influence the network’s predictions, helping to understand which features are important for the DQN’s decision-making process [18]. PolicyExplainer provides visualization of the states, policy, and expected rewards for an agent, helping users understand the rationale behind its actions [40].

In the field of robotics, in addition to the need for explanation of RL models, motion planning and control can be explained directly. Robotic tasks are often carried out in dynamic and complex environments, involving the real-time execution of multiple decisions. In these cases, the robot needs to be able to clearly explain its path planning and control decisions. Several methods are proposed to explain the control decision, MAPS-X explains by visualization plan as a short sequence of images representing a time period [34]. Natural language can also be used to answer behavior-related questions or directly provide explanations for the actions of agents [6, 19, 33, 48].

Despite progress in improving the explainability of RL models, their interpretation remains challenging in robotics due to the complexity of black-box models and the technical expertise required by current techniques, highlighting the need for intuitive, user-friendly tools to enhance understanding and enable wider adoption in real-world systems. Instead of relying on complex technical formulas, our approach focuses on providing intuitive insights, making RL models for robotics more accessible and comprehensible to practitioners.

2.3 Interactive Machine Learning Model Optimization

The most similar works to ours focus on visualization to support model optimization and debugging in RL, though not specifically for robotics. For example, DynamicsExplorer is a visual analytics tool designed for LSTM-based RL policies in robot control tasks, providing visualization to identify and address the failure cases [21], while DQNViz provides visualizations for agent behavior analysis and feature extraction with DQN models [55], and DRLIVE for RNN-based RL models [57].

While RL-focused visualization methods are limited, general visual systems have been developed to support analyzing and optimizing machine learning models. TensorBoard offers dataflow visualizations that aid the understanding of complex machine learning architectures [58], and platforms like Weights and Biases [7], Neptune [42], and ClearML [13] help track experiments and visualize training processes. Tools like Talaria visualizes model statistics interactively and simulates the impact on various metrics to optimize model performance [25].

Some systems focus on tracking and evaluating model behaviors, which is essential for identifying performance bottlenecks and edge cases. Zeno [11] is an interactive framework for assessing machine learning behavior, allowing users to visualize and test AI systems across different models and data types. SUMMIT [24] provides visualizations that highlight the features learned by a deep learning model and how these features interact to make predictions. Other systems assist in identifying and correcting model errors interactively [9, 59] or mitigating bias [3].

Although these general visualization platforms are of great help in analyzing complex models, they do not focus on the details of optimizing a specific model. To this end, there are some platforms offering model-specific visualizations, such as LSTMVis [51] for recurrent neural networks, GANViz [56] for Generative Adversarial Nets (GAN), XAutoML [62] for automated machine learning (AutoML) systems. In Convolutional Neural Networks (CNNs), CNNPruner focuses their structure and performance [35], while VAC-CNN is for comparison for image classification task [60], and VASS is for diagnosing semantic segmentation of movable objects [22]. For machine translation, tools like Angler [46], NMTVis [41], Neural Monkey [23], SoftAlignments [45] help analyze attention weights and improve translation understanding.

Existing machine learning visualization tools enhance model optimization but fail to address the unique challenges of RL in robotics, such as decision-making and dynamic movements, highlighting the need for specialized tools to meet the demands of robotic model optimization. Our approach specifically addresses these challenges by combining interactivity and visualization to better support practitioners in optimizing RL models for robotics.

3 Formative Study

In order to better understand the requirements for training locomotion policies and the challenges involved in both the training process and communicating the results, we conducted a formative study. We interviewed five domain experts (F1-F5) in the field of locomotion learning in a semi-structured manner. For more details on

the interviewees and procedures, please refer to the supplementary materials.

3.1 Current Practice

A typical locomotion learning training process is in a simulated environment where the robot performs actions, receives feedback from the environment, and updates its policy to maximize the rewards. The rewards term is one of the most important factors for locomotion learning. It encourages desired behaviors to achieve the learning objective. For example, rewards can be given for foot clearance (to avoid quick little steps) and forward velocity. On the other hand, penalties, or negative rewards, can be applied to discourage high joint torque (to reduce energy use) or jumping behavior.

Practically, there are two ways to evaluate the effectiveness of a trained policy: quantitatively, by assessing metrics like rewards and joint torque using tools like Matplotlib [26], or visually, through simulation tools like NVIDIA Isaac Gym [37] where experts can observe robot movements or other aspects like smoothness and foot clearance. Based on the evaluation, further refinements can be made iteratively by tweaking the reward function, adjusting the environment, or modifying the model, until the desired performance level is achieved.

After the objective is reached or a benchmark is met, the results must always be presented, whether to colleagues or in a formal paper. The easiest way to present these results is by showing plots generated in previous steps using Matplotlib or by taking screenshots from the simulation. However, these methods are not suitable for formal presentations. Typically, in academic papers, custom figures are created manually to highlight keyframes and illustrate important characteristics of the results. Moreover, videos can be created by compiling keyframes, providing a dynamic way to demonstrate the robot's performance over time.

3.2 Challenges

Based on the interviews, we identified three main challenges that the interviewees currently face when training, comparing, and communicating locomotion learning policies.

The gap between simulation and visualization. All interviewees mentioned the usage of NVIDIA Isaac Gym [37], which can provide a GPU-parallelized physics simulation environment for the efficient training, and render the movement of robot. However, it does not offer visualization for critical statistical metrics, such as the reward curve, that users find important. As a result, experts often rely on Python libraries such as TensorBoard [2] (used by 2/5 interviewees), Matplotlib [26] (3/5), and PlotJuggler [17] (1/5) to visualize these metrics. However, the challenge of using these libraries is the need for additional code to integrate the simulation tool with metric visualization. Even with this code in place, experts still need to view the statistics in a separate tool and manually align the performance in the simulation with the corresponding numbers in the visualization.

Displaying multiple policies. Comparison of different policies occurs at every stage of locomotion learning research. Visualization and judgment based on metrics such as action smoothness and environmental contact are crucial for this process. While simulation

tools allow users to run experiments and visualize multiple robots in parallel on a single GPU, comparing the movements of all the robots remains challenging. The main limitations are the burden of rendering and the difficulty of monitoring all robots simultaneously. Expert F5 mentioned using video logs for later comparison, but this approach was not flexible enough. This challenge is further compounded when training on a server without an interface for direct visualization. As Expert F2 mentioned, *“The main challenge I am facing is it is kind of hard to visualize how the policy is behaving when I am training in cluster.”*

Effort and time required to craft attractive displays. For casual communication, such as small group meetings with colleagues, a simple plot from TensorBoard or even just a reward value could be useful, especially since these colleagues have the expert knowledge to understand the data. However, these methods are insufficient for more formal situations, particularly when communicating with those outside the domain. As Expert F4 stated, *“Never show your TensorBoard plots on your report or your thesis, much less a paper.”* A well-crafted figure can effectively demonstrate movement, position, and other relevant information in a way that is both clear and comprehensive. Despite their utility, creating high-quality figures for academic papers can be time-consuming. Interviewees mentioned that they currently rely on tools like Adobe Photoshop and Premiere to create figures and videos, which often involves removing the background from each frame and manually concatenating them—a tedious and time-consuming process. As Expert F5 noted, *“If you publish a paper and you’re trying to make a figure for that paper. The figure can take a week.”* Despite these figures can be highly illustrative, the effort required to produce them discourages experts from using them regularly.

3.3 Design Requirements

Based on the key challenges, we summarize the following design requirements.

- R1 Flexible Robot Simulation:** The movement of the robots is the primary concern for practitioners, as it provides qualitative and intuitive insights into policy performance. In the formative study, participants mentioned using simulation tools to visualize robot movement and its interaction with the environment, enabling them to assess policy performance intuitively. Therefore, the platform should enable interactive and dynamic simulation of robots, allowing users to explore and manipulate the simulation flexibly. It should support seamless integration of simulation and analysis, reducing the need for switching tools and enabling intuitive, exploratory workflows.
- R2 Comprehensive Visualization of Key Metrics:** Practitioners often struggle with evaluating raw numerical data, such as reward metrics, requiring additional effort to generate visualizations. The platform should provide customizable and integrated visual representations of critical metrics, enabling users to correlate statistical information with observed behaviors directly. Visualizations should facilitate multi-dimensional comparisons (e.g., time-series, spatial trajectories, rewards) and align with the cognitive needs of users for both analysis and communication.

R3 Easy Multi-Policy Comparison: Comparing different policies is a core task during the whole locomotion training and refinement process. The platform should allow iterative, direct, and easy-to-understand comparisons of policy performance across multiple dimensions (e.g., behaviors, trajectories, rewards), making it easier to fine-tune and improve policy performance. The interface should be designed for intuitive understanding, minimizing cognitive overhead and enabling rapid exploration of alternatives.

R4 Support Convenient Visual Report Communication: Preparing visual reports for presentations and publications can be tedious and time-consuming. The platform should offer tools for automated and customizable visual report generation, such as robot images, trajectory snapshots, and comparative charts. This would enable practitioners to prepare and present their results efficiently.

4 Content Analysis

To fulfill **R4**, and gain deeper insights into the types of visual aids commonly used in robotics and locomotion training, we conducted a content analysis of existing papers in the fields of robotics and locomotion training. We chose to analyze the visual information in these publications because they are often meticulously crafted to demonstrate effectiveness or compare different approaches. The findings from this analysis directly informed the design of MRISA, ensuring its alignment with established practices.

4.1 Method

4.1.1 Data Sampling. A senior robotics researcher from our team curated a selection of recent publications in the field of robotics, focusing on those that center around locomotion training and contain a substantial amount of visual content. The final list (detailed in supplementary materials) comprises 20 papers from leading robotics journals and conferences, including Science Robotics, International Journal of Robotics Research (IJRR), IEEE Transactions on Robotics (TRO), Robotics: Science and Systems (RSS), and ICLR. We then extracted relevant figures from these papers for further analysis. Our selection focused on figures that specifically depict robot behavior or evaluate the robot’s performance, while excluding those related to algorithm workflows or model diagrams. To avoid redundancy, we included only one instance of similar figure types from each paper. Additionally, we included both the selected figures and the corresponding captions, ensuring that the context were included. By following this process, we compiled a dataset of 58 unique figures.

4.1.2 Thematic Analysis. Two HCI researchers from our team independently reviewed a pilot sample of seven papers, carefully examining and coding the figures to develop an initial coding scheme. This iterative approach ensured that the coding scheme captured a broad range of visual strategies used in the field. The two researchers then met to discuss discrepancies, refine the coding scheme, and identify emerging themes based on shared patterns and salient differences in the visualizations. Using the refined coding list, we proceeded to analyze the remaining figures and papers and continued to revisit and update our themes as new data emerged to ensure all findings were grounded in material. To manage the coding process and track

iterations, we used a spreadsheet tool for managing and organizing the codes and themes.

4.2 Analysis Result

Our final categorization of locomotion policy presentation strategies consists of four main categories: data content, data presentation, robot presentation, and the robot's background. We summarize themes within these categories as a design space and sort them in descending order to highlight the most frequently used patterns, as shown in Figure 3. We have incorporated these insights into our visualization design, implementing many of the dominant visual presentations and aids identified in this analysis, while also carefully considering the specific robot type and striving for consistency.

4.2.1 Data Content. The data content used in robotics research papers reflects various analytical focuses, but the overarching goal is to support the assessment of movement and control. Transition/position data, the most frequently used type (11 out of 20 papers), provides information on the robot's spatial location, a foundational aspect for any locomotion task, as it enables tracking of the robot's path through an environment. Scalar metrics such as reward terms, are also commonly employed (10/20), which can further quantify policy characteristics. Rotation and orientation data (8/20) contribute additional detail on the robot's orientation in three-dimensional space, a crucial factor in aerial applications such as drones. Velocity/speed data (8/20) and torque data (6/20), provide vector information, offering insights into both magnitude and direction. Lastly, transformation / pose data (5/20) reflects the robot's body configuration, essential for complex tasks requiring precise positioning.

4.2.2 Data Presentation. All papers in our content analysis utilize at least one form of mathematical abstraction (e.g., graphs or charts) to visualize and clarify their data. Given that most studies employ time-series data, line charts emerge as the predominant utilized format (16/20). Additional plot types are also used to represent data distributions and categorical comparisons.

Keyframes are also frequently used (15/20), either as single overlapping images or as multiple subfigures, to directly illustrate transformations or pose changes. Trajectory (12/20) is commonly used to represent translation or position changes, allowing readers to trace movement paths over time.

4.2.3 Robot Presentation and Background. The presentation of robots and their backgrounds is often coordinated based on the robot's unique features and tasks. Both real-world photos (13/20) and simulation models (12/20) are widely used, which can be within the same paper. For instance, simulation figures may be used initially to illustrate the training process, followed by real-world photos to present the robot's performance in real-world scenarios. Accordingly, the backgrounds vary significantly depending on the robot presentations and tasks.

5 MRISA: A Multi-Robot Interactive Simulation and Analysis Platform

MRISA is designed to address the challenges practitioners face throughout the locomotion learning process, as identified in our formative study, by incorporating visual aids identified through

content analysis. The goal is to empower practitioners with a more efficient and user-friendly environment for conducting high-quality locomotion studies, reducing overhead and allowing them to focus on the core aspects of their research.

5.1 Visualization Design

To meet the identified design requirements, we design a range of functionalities and group them by panels in the MRISA interface (Figure 1). Each function is designed to streamline the training process, addressing key challenges faced by practitioners, and supporting both qualitative and quantitative evaluations of multiple policies.

5.1.1 Policy Simulation. As discussed in Section 4.2.1, the most common data content during locomotion training is the movement data of robots, including their position and rotation, which provides an intuitive, qualitative understanding of a policy's performance. According to experts in the formative study, practitioners often rely on simulation tools to observe robot movement as part of the training workflow. To support this, MRISA allows users to load pre-collected policy logs for multiple robots individually through Top Control Panel, while the Simulation Panel generates interactive simulations based on the loaded data (R1).

Interactive features enable users to explore a robot's behavior and focus on specific aspects of interest. The Simulation Panel provides options to start, pause, stop, or step through simulations, offering precise control over playback. Additionally, users can directly interact with the robot by manually adjusting its joints. Robots can either be locked at the origin or allowed to move based on the loaded data. Locking robots at the origin simplifies comparisons by highlighting differences between policies, whereas allowing movement showcases their actual behavior. To facilitate comparisons, users can highlight a selected robot or make it invisible. If a robot is no longer needed, it can be deleted from the platform.

5.1.2 Visual Data Inspection. In the formative study, experts noted that practitioners often use metrics to evaluate policies quantitatively. To facilitate different types of insights, MRISA supports line charts, which display continuous changes and trends—recognized as the most common data presentation method in Section 4.2.2—and heatmaps, which effectively highlight abnormalities. The Global Time-based Data View and Local Data View (Figure 4) display scalar metrics, such as reward terms, joint position, velocity, and torque, as analyzed in Section 4.2.1, enabling accurate quantitative evaluation (R2). Global Time-based Data View provides an overview throughout the entire simulation, allowing users to maintain a perspective and quickly identify obvious problems.

Plots can be grouped by robot or by specific metrics, such as total rewards, enabling users to compare data across robots within a single visualization (R3). To enhance clarity, selected options can be highlighted for emphasis. Additionally, when grouping by robot and analyzing joint-related metrics, hovering over a joint in Simulation Panel highlights the corresponding line in red, providing intuitive and interactive feedback.

5.1.3 Subtle Movement Tracking. As discussed in 4.2.2, trajectories are a commonly used data presentation method. They are especially valuable for visualizing subtle movements that may be difficult to

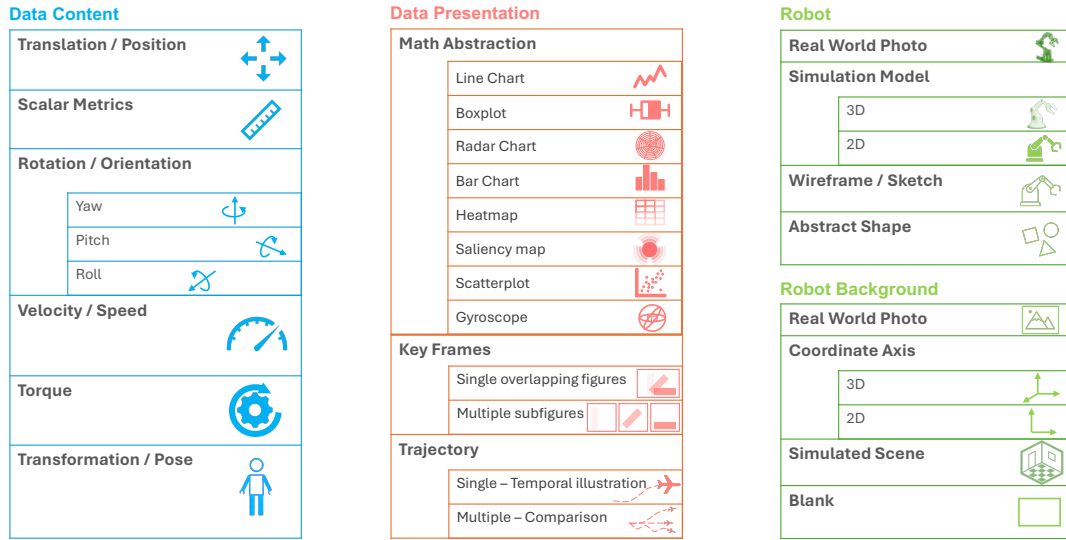
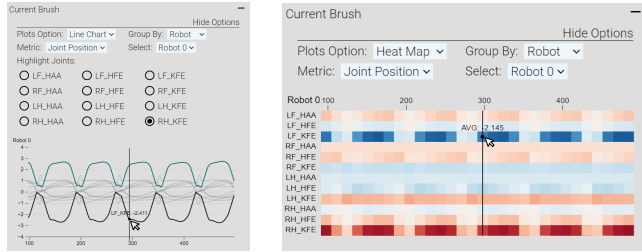


Figure 3: A design space for locomotion policy presentation strategies based on the literature review result. Within each category, we organize the strategies in descending order according to the number of papers that employ each strategy.



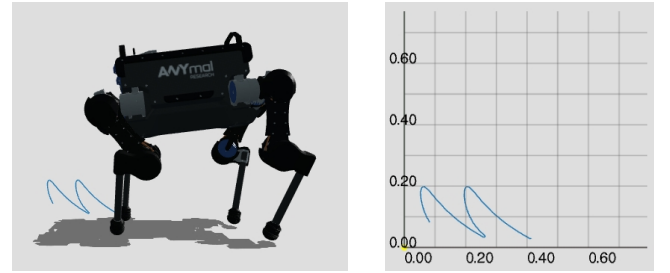
(a) Line chart in Local Data View

(b) Heatmap in Local Data View

Figure 4: The Local Data View within the selected time range, chosen using brush from the Global Time-based Data View: (a) Line Chart showing joint positions for Robot 0, highlighting the LH_HFE joint (green) with a hover on the RH_KFE joint (black). Other joints are shown in light grey. (b) Heatmap representing the same data, with a hover on the LF_KFE joint to display the corresponding value in the grid.

discern directly on a robot. For example, foot clearance is important for the quadruped robot, as the robot should not take quick small steps. As shown in Figure 5a, visualizing the trajectory of the foot makes the robot's step patterns clearer. The trajectory can be added flexibly into the Simulation Panel with any point on the joints (R1), with colors selected from a palette of 10 distinct hues based on the robot index.

When the robot moves, the trajectory of the selected point is continuously displayed, providing a visual representation of the movement path. To maintain consistency, trajectories are automatically applied to the same corresponding joint points on all robots



(a) Trajectory in Simulation Panel

(b) Trajectory in Trajectory Coordinate Panel

Figure 5: The trajectory of the robot's foot during quick small steps as indicated by the blue line. The robot is walking in a small-step gait with a stride length of less than 0.2 meters and a height of under 0.2 meters.

in the simulation. Analyzing these trajectories can reveal variations in movement caused by differences in robot design, control algorithms, or environmental conditions (R3).

For more detailed trajectory evaluations (R2), the Trajectory Coordinate Panel (Figure 5b) includes an axis indicator with scales, enabling coordinate-based comparisons of robot trajectories, and further supports accurate cross-robot performance analysis (R3).

In addition, the Local Data View focuses on a selected period using the brush tool from Global Time-based Data View, providing a more detailed and precise analysis of the data within that timeframe (R2). If the brush is locked, the simulation repeatedly runs only within the selected period, allowing users to focus specifically on the movement.

5.1.4 Result Presentation. As discussed in Section 4.2.2, keyframes are a widely used method for presenting data. Based on insights

from the formative study, roboticists often manually capture screenshots from simulation tools and use external software to extract the robot from the images, removing the background for a cleaner presentation. *MRISA* provides a simpler solution: users can capture keyframes directly from Simulation Panel by clicking a button. This feature generates snapshots of the current simulation step with a transparent background, which can be easily downloaded or copied for use in presentations or publications (R4). This feature is particularly useful when simulation training is conducted on a server, which often lacks a user interface, making it difficult to take real-time screenshots.

Additionally, the plots from Global Time-based Data View, Local Data View and Trajectory Coordinate Panel provide easily accessible data visualization results (R4). These plots can be customized to highlight key metrics, trends, and performance indicators, allowing users to easily show the results of comparing and contrasting different policies.

5.2 Development Details

This platform is implemented as a web application using JavaScript, with Three.js [10] for 3D rendering and D3.js [8] for data visualization. This work builds upon the URDF Loaders framework, an open-source project under the Apache License 2.0 [31]. This framework enables the loading and rendering of a robot using Unified Robot Description Format (URDF) models within a web environment via Three.js, and supports some basic interactivity. The URDF model is designed as human-readable, including the details of the robot's structure, dynamic parameters, collision geometries, and more. It is widely supported across simulation platforms, and an aggregated open-source dataset is available that includes more than 195 different robot models, covering most of the robots commonly used in both research and industry [53]. *MRISA* currently uses a URDF model from ANYmal [27].

6 Usage Case Scenario

Imagine Bob, a robotics engineer, who is working on tuning a robot control policy for better foot clearance. His goal is to ensure the robot lifts its legs higher and more distinctly, avoiding small, quick steps, to achieve smoother and more effective movement.

Single policy examination. Bob starts by training a policy based on his prior experience and uses a Cloud GPU server to accelerate the training process. He loads the policy logs into *MRISA*, Bob directs his attention to the Simulation Panel (Figure 6.1), and notices the left hindleg appears abnormal, indicating a potential flaw in the control policy. To investigate, Bob turns to the Global Time-based Data View (Figure 6.2) to view performance metrics over time, including joint torques and reward terms. Using the brush tool, Bob selects a time window to examine the movement pattern and checks the heatmap in the Local Data View for any abnormal data (Figure 6.3). He located the excessive torque in the left hip joint (LH_HAA) as the cause of the issue.

Multiple policies comparison. After identifying issues with the previous policy, Bob refines the reward function and trains three new policies, each incorporating different adjustments to the reward function and control parameters. To track the movement of the left hindleg, Bob marks a tracking point on it and generates

the trajectories for all robots (Figure 7.2). The trajectories reveal the nuanced effects of each policy on the robot's movement patterns. With more analysis of cumulative rewards, joint torques, and velocities across the policies in Local Data View (Figure 7.3), he identifies which reward modifications improve performance.

Formal presentation. Following the comprehensive analysis of multiple policies, Bob prepares a formal presentation of his findings to his team. To support his analysis, Bob presents a figure that visually combines the robot's position and movement at different moments in time, effectively illustrating the movement sequence. This figure is like a multiple-exposure photograph, but with keyframes captured in the platform. Bob overlays keyframes with a backdrop to produce a clear visual representation of the robot's motion (Figure 8). This process only takes him 2 minutes, significantly faster than Figure 2. Next, he presents plots of the reward terms to highlight which modifications are most effective and how they contribute to the overall improvement in the robot's gait. Bob discusses the strengths and weaknesses of each approach, providing a roadmap for future research and further policy refinement.

7 User Evaluation

To gain feedback about the usability and usefulness of *MRISA*, we conducted a user study with 14 domain practitioners from Europe, Asia, and North America (1 female, 13 males, U1-U14). The gender imbalance among participants mirrors the broader gender gap in the robotics field. Our primary focus was on observing user behaviors and collecting feedback, which allowed us to improve *MRISA* based on the participants' suggestions. In this study, participants were required to complete two tasks to evaluate given locomotion learning policies. The purpose of these tasks was not to assess the participants' abilities but to provide them with an opportunity and scenario to experience *MRISA*. Following these tasks, participants provided quantitative ratings and qualitative feedback through a questionnaire and interviews. For more details on the study design, please refer to the supplementary materials.

7.1 Results

Overall, all participants rated the framework positively in the questionnaire (Figure 9) and were satisfied with the experience using the framework (12/14 strongly agree, 2/14 agree). Participants also provided non-negative feedback for all the features they used (Figure 9), while the "Not Applicable" category was for features they did not try.

The framework efficiently assists users in analyzing and comparing policies. All participants agreed the framework allowed them to quickly achieve the goals for tasks (for Task 1, 14/14 strongly agree; for Task 2, 11/14 strongly agree, 2/14 agree and 1/14 somewhat agree). The accuracy of Task 1 was 100% and the average accuracy rate of Task 2 was 80%. Notably, U6 failed in Subtasks Q2.2 and Q2.3, by choosing the robot with more torque consumption and shaking instead of less, and only agreed (rather than strongly agreed) with the system's effectiveness in Task 2. Similarly, U5 and U7 failed to find the correct reward term in Subtask Q2.3, likely due to a limited understanding of the system, as they agreed, rather than strongly agreed, with its clarity and ease of use.

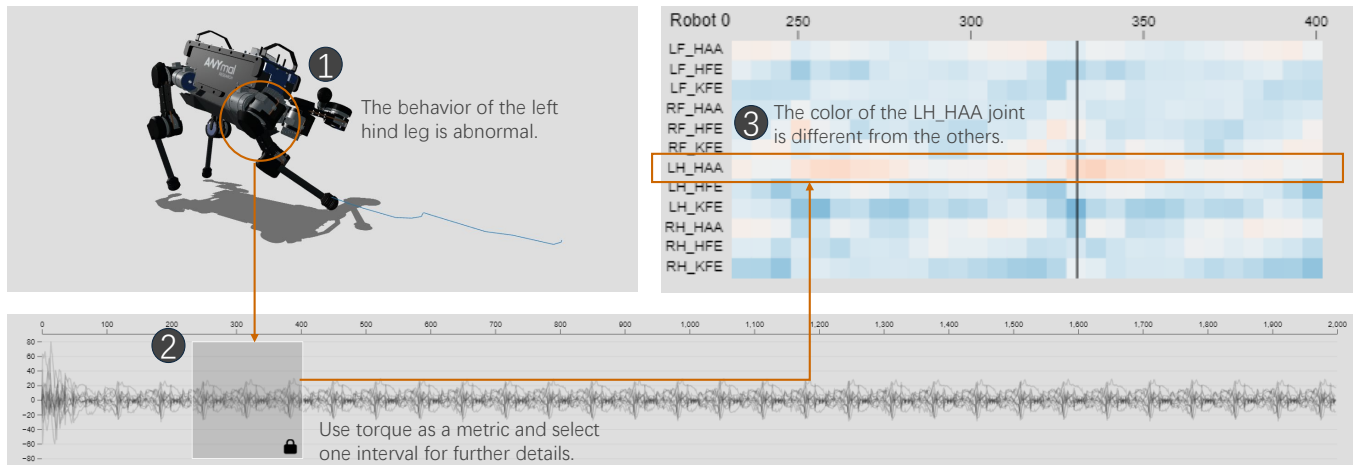


Figure 6: Single policy examination. (1) The robot moves within the Simulation Panel, where abnormal behavior is identified in the left hindleg. (2) The Global Time-based Data View displays a line chart of torque value for all 12 joints over time. (3) The Local Data View presents a heatmap of the selected interval, highlighting prominent positive torque (red) in LH_HAA, while other joints predominantly show negative torque (blue).

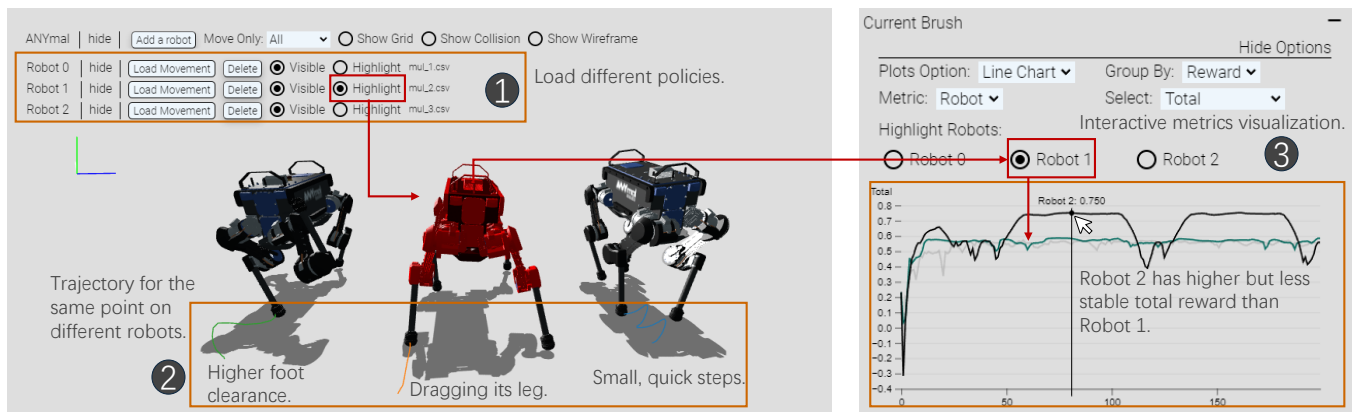


Figure 7: Multiple policies comparison. (1) Policies for multiple robots are loaded and managed in Top Control Panel. (2) The Simulation Panel visualizes the trajectory of a common point on different robots, enabling direct comparison. The trajectory reveals distinct behaviors: the left robot lifts its leg, the middle robot drags, and the right robot takes small, quick steps. (3) The Local Data View presents a line chart of performance metrics (e.g., total reward) for selected robots, showing Robot 2 (black) with a higher but more dynamic total reward compared to Robot 1 (green). Red lines highlight the selected robot (Robot 1) for focused analysis.

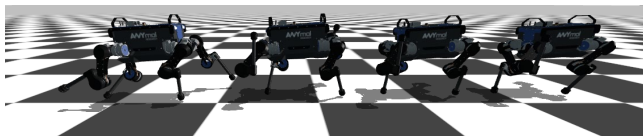


Figure 8: Formal presentation. The figure consists of four keyframe snapshots captured in MRISA.

Distinct patterns emerged in system usage based on task types. For Subtask Q1.1, which required the value of total reward, all participants consulted plots in Global Time-based Data View or Local

Data View. In contrast, Subtasks Q1.3 and Q2.4 were open-ended, leading to varying behaviors among participants. For Subtask Q1.3, U1 relied on previous findings, U6 used only Global Time-based Data View, U5 and U11 focused on simulation results, while others combined data view and simulation. For Subtask Q2.4, all participants used simulations, with six also incorporating data views. Subtask Q2.1 allowed both methods and also showed varying strategies.

Participants highlighted the framework's potential, with U7 noting its convenience for lectures due to the web-based interface, which eliminates the need for local installation. U1, U8, and U9 expressed a strong willingness to integrate it into future work. For

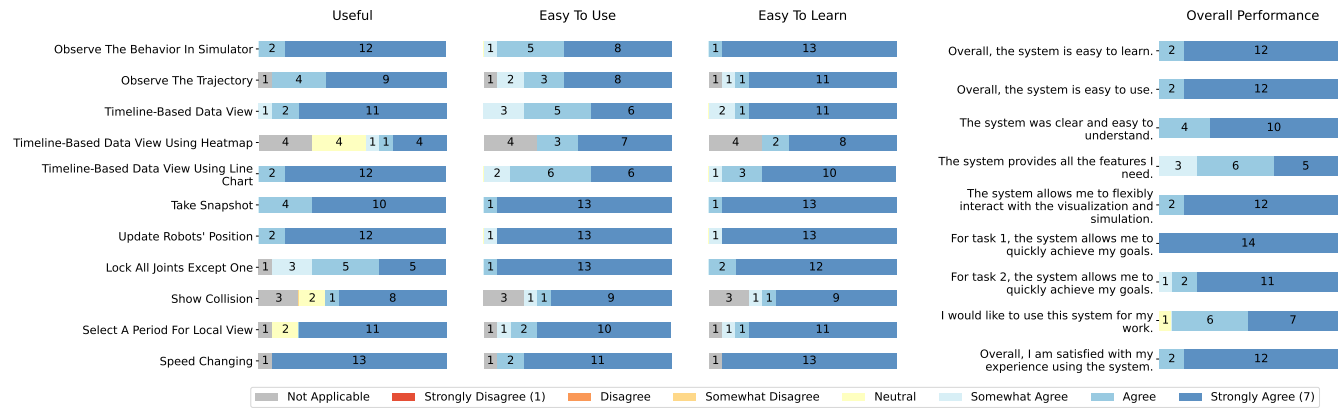


Figure 9: The user feedback on each feature and on the overall satisfaction using a 7-point Likert scale ranging from "Strongly Disagree (1)" to "Strongly Agree (7)". A majority of participants expressed strong agreement across most questions, indicating high levels of user satisfaction with the framework.

learning, U12 remarked that a 5-minute tutorial suffices. Overall feedback was positive, but insights on individual features provided nuanced perspectives on their contributions.

Simulating policies helps to form an intuitive understanding of the behavior. The simulation feature emerged as the most frequently used tool in the study, with all participants utilizing it to observe robot behavior. Most participants (12/14) strongly agreed, and the remainder (2/14) agreed, that the "Observe The Behavior In Simulator" function was useful. Participant U9 highlighted its effectiveness in policy evaluation, explaining how the simulation helps pinpoint issues, which can then be cross-verified using plots in Global Time-based Data View and Local Data View. Similarly, U11 emphasized its utility for debugging. For potential applications, U1 noted that the platform uniquely enables robot visualization post-server training, a capability lacking in other tools. However, a few participants (U2, U5, U8) found tracking robot movements challenging and suggested implementing an auto-follow camera to update positions dynamically. Additionally, some participants (U2, U3, U7, U12) expressed dissatisfaction with the single-joint movement feature, recommending an enhancement to allow movement of selected joints for visualizing entire leg motions.

Timeline-based Data View for numerical metrics. The plots in Global Time-based Data View and Local Data View were primarily used for metrics analysis and policy comparison. All participants relied on these features for Subtasks Q1.1 and Q2.2, while in Subtask Q2.3, only U9 opted to use the simulation instead. Both Global Time-based Data View and Local Data View offered heatmap and line chart options. Participants found the data view useful overall (11/14 strongly agree, 2/14 agree, 1/14 somewhat agree). Line charts were favored, receiving higher ratings and being widely used, while heatmaps were used less frequently and rated lower, which also corresponds to the content analysis in Section 4.2.2. U13 and U14 highlighted the unique advantage of having an integrated data view, as other tools require manual Python scripting for numerical data. U11 noted that line charts are valuable for observing gait patterns, while U5 and U3 emphasized the utility of heatmaps for style quantification and gait mapping, respectively. Some participants

identified areas for improvement. U1 and U6 found the abundance of options in the data view overwhelming for first-time users, with less experienced participants struggling to interpret values. U13 suggested allowing customizable line colors for different robots, while U2, U8, and U12 recommended a y-axis zoom function for line charts to enable more detailed analysis.

Snapshot keyframes simplify reporting. The "Snapshot a Keyframe" feature was strongly agreed upon by all participants as being easy to use and easy to learn. It provided a straightforward method for capturing snapshots with transparent background (Figure 10 as an example). U12 strongly agreed on the feature's usefulness, stating, "If I want to publish a paper one day, I think this snapshot can be very useful." Participant U10 mentioned that, in current practice, it is most common to screenshot the simulator directly. Additionally, U10 noted that using Blender requires adjusting the light source, making the screenshot process very time-consuming.

Trajectory to enhance visualization. The trajectory feature was not directly required for the tasks, resulting in less frequent use than anticipated. However, participants frequently mentioned its value after experiencing it during the tutorial. U11 used the trajectory in Subtask Q1.3 to align the reward plot with the simulation. Participants highlighted the complementary role of the trajectory feature. U2 noted it helps assess the aesthetic quality of movements, which is common practice in such analysis, while U7 found it particularly useful for observing specific actions or repeated patterns. U13 praised its convenience, stating, "I haven't found another tool that is so convenient, and it's great that I don't have to write code myself." U3 added that the feature is beneficial for tracking humanoid robot knuckle movements. U14 also commented, "Trajectory is more effective for illustrating movement in a static format, such as in a paper." While the feature was generally appreciated, participants tended to focus more on the trajectories displayed directly on the robots rather than those shown in the Trajectory Coordinate Panel. U10 complained the trajectory in Trajectory Coordinate Panel was too small to observe clearly, and U2 felt the panel was unnecessary, as the trajectory provided more qualitative insights than quantitative.



Figure 10: The keyframe captured by U6 during Subtask 1.2. The robot lifts its left foreleg to an abnormal height, which is proof of its unnatural movement.

8 Discussion and Future Work

In this section, we begin by exploring the potential for generalizing the system’s design features to other applications. We then address the current limitations of the system and outline possible directions for future enhancements.

8.1 Visual Designs for Mixed-Methods Analysis

The design requirements proposed in this study demonstrate potential for generalization across diverse domains beyond robotics where simulation, visualization, and comparison are integral to analysis and decision-making.

Interactive Simulation. The capability to simulate robots flexibly is crucial for supporting various types of robots and robotic systems, extending beyond just the policy training process. Simulation serves as an abstraction of the real world, and flexible simulation enables a deeper understanding of robot movements and postures, aiding in both analysis and design across diverse robotic applications [12, 14].

Multi-Dimensional and Customizable Visualizations. Customizable visualizations that span multiple dimensions are valuable for enhancing the explainability and interpretability of other machine learning models, especially in domains where complex data relationships must be communicated effectively [21, 51, 60].

Simplified Model Comparison. Simplified and intuitive tools [2, 17, 42] for comparing machine learning models can accelerate the training process and improve optimization efficiency. This comparison not only streamlines the evaluation of different models but also enhances the decision-making process by making performance metrics more accessible and interpretable.

Keyframes for Key Objects or Events. Features like keyframe snapshots can extend beyond robotics to any domain involving time-series data, where representative objects or moments are crucial for analysis and communication. For instance, in video-related fields like video summarization [38], keyframes offer an efficient method to capture and highlight pivotal moments, simplifying the process of understanding and conveying complex video sequences.

8.2 Limitations and Future Work

The user evaluation provided valuable insights for improving *MRISA* by enhancing its usability and functionality. In this section, we will discuss the identified limitations and potential directions for future development.

Beyond a single condition study. Our user evaluation was conducted using a single-condition study, as there is no existing tool that offers an equivalent feature set for a direct baseline comparison. While this prevents a direct performance benchmark, it enabled us to focus on understanding how *MRISA* fits into practitioners’ workflows and what unique benefits it provides. Future work could consider controlled experiments with a fair designed baseline condition to more accurately verify the effects of system improvement. For example, users could be asked to perform the same task using both *MRISA* and the traditional method they are accustomed to.

Expanded customization capabilities. Currently, *MRISA* supports simulations for a quadruped robot in a flat ground environment, which is a significant limitation. Future updates will include support for loading URDF files, enabling users to simulate their custom robot models. Additionally, we plan to integrate features for loading and customizing environmental setups, allowing simulations in more varied and complex scenarios.

Enhanced visualization features. While *MRISA* provides trajectory simulations and metric visualizations to deliver both quantitative and qualitative insights for policy comparison, there is room for further enrichment. Planned improvements include visual feedback for robot-environment interactions, such as contact points with the ground or obstacles, displaying the center of mass for each joint, and adding local axis indicators to enhance the understanding of joint dynamics. These enhancements aim to provide users with a more comprehensive comparison of policies across multiple dimensions.

Additional result presentation formats. Although keyframes and metric visualizations effectively support practitioners in communicating their findings, offering additional result formats could further improve communication. For example, we plan to implement features to export simulation videos with overlays such as trajectories and speed metrics, providing dynamic visualizations of policy performance.

Further deployment study. The current user evaluation focused on observing user behaviors and gathering feedback during short-term interactions. Future work will include deployment studies to assess the long-term effectiveness of *MRISA* in real-world environments. These studies will aim to cover diverse usage scenarios and setups, uncovering additional user requirements and identifying issues that may not arise during our user evaluation.

9 Conclusion

In this work, we present *MRISA*, an interactive platform that supports analyzing and comparing multiple policies with both robot simulation and data visualization to support more convenient policy training and result communication. *MRISA* was built upon insights from a formative study and content analysis which identify the key challenges and the prevalent representations in the field, addressing key gaps in existing tools. The positive feedback from our user evaluation demonstrates *MRISA*’s potential to significantly improve the efficiency of the policy training process, making it easier for practitioners to refine and present their work. *MRISA* has the capability to serve as a comprehensive platform for locomotion learning, offering more effective and user-friendly assistance

for understanding and optimizing locomotion policies. We plan to open-source this tool to further contribute to the development of analysis tools within the robotics community. By providing an intuitive and efficient way to visualize and compare policies, we aim to lower the barrier to entry for locomotion training, encouraging interdisciplinary innovation and collaboration in the field.

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A Formative Study Interviewees and Procedure

We conducted interviews with five domain experts (F1–F5, all male), with an average of 3.2 years of experience in training locomotion policies, specifically for quadruped robots. Four experts are from academia, and one is from industry, providing both theoretical and practical perspectives. Although RL for locomotion has recently gained significant attention, it should be noted that the field is still relatively new, making three years of experience notable. All interviewees are actively working in the field.

The interviews began with background questions about the interviewees’ professional experience in locomotion learning. Subsequently, they were asked about various aspects of their work, including single policy training, comparing different policies, and communicating results, with a focus on the tools they used and the challenges they faced.

B Paper List in Content Analysis

C User Evaluation

C.1 Participants

We recruited 14 participants (1 female, 13 males) with specialized experience in robotics, particularly in the domain of locomotion learning. As a side note, the unbalanced gender distribution is a reflection of the existing gender disparity in the field of robotics. To further enhance our participant pool, we supplemented convenience sampling with snowball sampling. After initial recruitment from our known contacts, we asked participants to refer other qualified colleagues in the field, allowing us to reach a wider audience of experts in locomotion learning. On average, participants had three years of experience in this emerging field. Geographically, they were from different regions: one from North America, ten from Europe, and three from Asia. Educationally, the group comprised 4 PhD students and 10 Master’s students in Robotics. Two participants requested to conduct the study in person, while the remaining participants completed the study remotely via Zoom. Each participant received an approximately 30 USD gift card in their local concurrency as compensation for one hour of their time. All participants were informed of the study’s purpose and provided consent before participation. Data were anonymized to ensure participant privacy, and they were allowed to withdraw from the study at any time without penalty. This study was approved by our institutional ethics commission.

ID	Paper Title	Venue	Year
T1	Learning Agile and Dynamic Motor Skills for Legged Robots	Science Robotics	2019
T2	Learning Quadrupedal Locomotion over Challenging Terrain	Science Robotics	2020
T3	Reaching the Limit in Autonomous Racing: Optimal Control versus Reinforcement Learning	Science Robotics	2023
T4	Trajectory Generation and Tracking Control for Aggressive Tail-Sitter Flights	IJRR	2023
T5	Representation-Free Model Predictive Control for Dynamic Motions in Quadrupeds	TRO	2020
T6	Real-world humanoid locomotion with reinforcement learning	Science Robotics	2024
T7	WoCoCo: Learning Whole-Body Humanoid Control with Sequential Contacts	Arxiv	2024
T8	Rethinking Robustness Assessment: Adversarial Attacks on Learning-based Quadrupedal Locomotion Controllers	RSS	2024
T9	Learning quadrupedal locomotion on deformable terrain	Science Robotics	2023
T10	Learning robust autonomous navigation and locomotion for wheeled-legged robots	Science Robotics	2024
T11	Scientific exploration of challenging planetary analog environments with a team of legged robots	Science Robotics	2024
T12	Versatile multicontact planning and control for legged loco-manipulation	Science Robotics	2023
T13	BirdBot achieves energy-efficient gait with minimal control using avian-inspired leg clutching	Science Robotics	2022
T14	Multi-expert learning of adaptive legged locomotion	Science Robotics	2020
T15	AntBot: A six-legged walking robot able to home like desert ants in outdoor environments	Science Robotics	2015
T16	Design principles of a human mimetic humanoid: Humanoid platform to study human intelligence and internal body system	Science Robotics	2017
T17	ANYmal parkour: Learning agile navigation for quadrupedal robots	Science Robotics	2023
T18	Agile and versatile climbing on ferromagnetic surfaces with a quadrupedal robot	Science Robotics	2022
T19	Inverted and vertical climbing of a quadrupedal microrobot using electroadhesion	Science Robotics	2018
T20	FLD: Fourier Latent Dynamics for Structured Motion Representation and Learning	ICLR	2024

Table 1: Paper Lists in Content Analysis

C.2 Study Protocol

The study began with a 15-minute tutorial that included three warm-up tasks designed to familiarize participants with the platform’s features. During the tutorial, the functionalities were demonstrated to the participants with illustrative figures and GIFs. The warm-up tasks covered the most important features, including interacting with time-view data plots, adding trajectories to simulations, and capturing snapshots for keyframes.

Following the tutorial, participants were asked to complete two tasks with *MRISA* with screen sharing, each lasting approximately 15 minutes. These tasks were designed in a scenario that evaluated policies trained by a hypothetical undergraduate student, with the policy data collected using *RaiSim* [28].

In the first task, participants analyzed a single policy and provided feedback and suggestions for improvement to the hypothetical student. To become acquainted with the plots in Global Time-based Data View and Local Data View, participants first identified the maximum value of the total reward (Question Q1.1). They then judged whether the robot’s movement was neutral and explained their observations by capturing keyframes (Q1.2). For the final question in Task 1 (Q1.3), participants were asked which reward terms could be adjusted to enhance the robot’s performance, along with their reasoning.

In the second task, we presented participants with three policies that the hypothetical student created based on the suggestion, asking participants to compare them. Participants first compared the foot clearance of the robots (Q2.1), which could be done by Simulation Panel and Trajectory Coordinate Panel. They then compared the shaking and torque magnitudes of the three robots using plots (Q2.2 and Q2.3). Lastly, participants assessed which of the three control policies demonstrated the best performance, providing justifications for their choice in an open-ended question (Q2.4). These tasks are designed to reflect common challenges that robotics experts face when training control policies using RL algorithms.

The study concluded with a feedback questionnaire and a post-study interview, which together took approximately 15 minutes. The questionnaire assessed the usefulness and usability of features and the system as a whole, using a 7-point Likert scale. The interview gathered qualitative feedback on the system and suggestions for future improvements.

C.3 Data Collection and Analysis

In this study, we collected both quantitative and qualitative data.

Quantitative data included responses from a 7-point Likert scale questionnaire and close-ended questions embedded within the tasks. The questionnaire assessed each feature based on three key aspects:

usefulness, ease of use, and ease of learning. In addition to evaluating individual features, the overall performance of the system was also assessed. The close-ended questions within the tasks provided data on the success rate for each task, showing correlations between subtasks and specific features.

Qualitative data were obtained through open-ended questions (Q1.3 and Q2.4) and semi-structured post-task interviews, allowing participants to express criticisms, offer suggestions, and discuss the

usability of related features. We also recorded feature usage logs to analyze frequency and patterns of use across different subtasks or purposes.

Through these analyses, we obtained a comprehensive view of how the system's features were utilized, their impact on task performance, and areas for future improvement.

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Table 2: Participant Background Information in User Evaluation

Participant ID	Years in Robotics	Gender	Education Level	Geographical Region	Robot Type
U1	3	Male	PhD	North America	Manipulator
U2	5	Male	Master	Europe	Quadruped robots and Humanoid robots
U3	6	Male	Master	Europe	No Particular
U4	2	Male	Master	Europe	No Particular
U5	5	Male	PhD	Europe	Humanoid robots
U6	3	Male	Master	Europe	Robotics manufacture
U7	3	Female	Master	Europe	Robotic arms
U8	1	Male	Master	Europe	Quadruped robots
U9	1	Male	PhD	Asia	Robotic arms and Humanoid robots
U10	2	Male	PhD	Asia	Robotic arms
U11	4	Male	Master	Europe	No Particular
U12	1	Male	Master	Asia	Quadruped robots
U13	2	Male	Master	Europe	Quadruped robots
U14	3	Male	Master	Europe	Quadruped robots